

Section 1

Microsoft Azure AZ900

A massive collection of internet-based services for building, deploying, and managing applications without owning physical servers. Which called cloud computing such as AWS, GCP, and Azure.

Azure	For enterprise applications
AWS	For machine learning workloads
Google Cloud	For data analytics



Questions on the exam

Domain 1:

Describe cloud concepts (25-30%)

Domain 2:

Describe Azure architecture and services (35-40%)

Domain 3:

Describe Azure management and governance (30-35%)

Minimum scores

Scale: 100 to 1000

Passing score: **700**

Orientation & Quick Win	Cloud Computing Concepts
Azure Architecture	Azure Compute Options
Azure Networking Services	Azure Storage Services
Identity, Access & Security	Governance & Cost Management
Deploy, Monitor, Repeat	

Azure Administrator

1. Log into Azure at <https://portal.azure.com>
2. Understand the Azure resources hierarchy: Management Group -> subscription -> resources group -> resource.
3. Your subscription are your billing and management boundary.
4. Resources groups are logical containers for organizing related resources.
 - Tags are not inherited from resource groups to resources
 - If you delete a resourcing group, what happens? All resources inside are deleted and there is not undo button.
 - The left navigation bar gives you a quick access to azure services.

Section 2

Cloud Computing

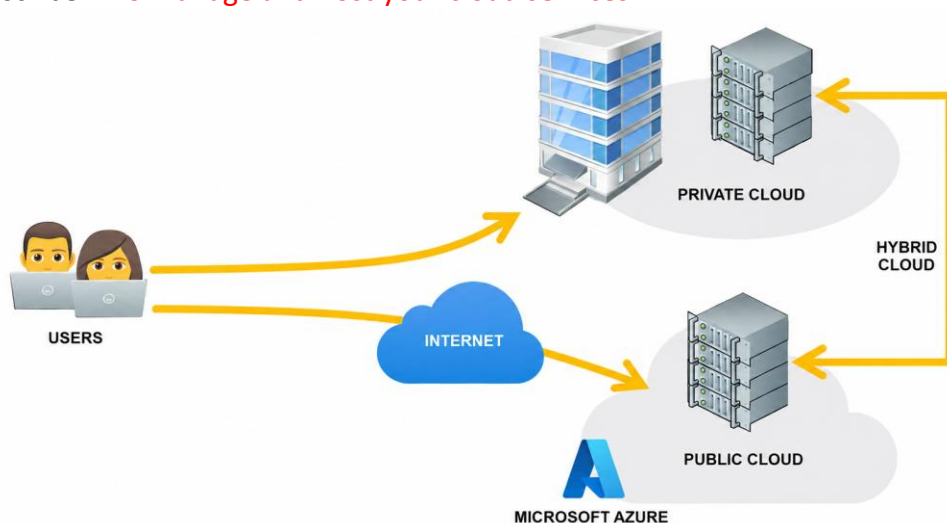
The delivery of **computing services over the internet** or “the cloud”.

Computing services: (6)

- **Servers**
- **Storage**
- **Databases**
- **Networking**
- **Software**
- **Analytics**

Cloud Models 3

Describe **who manage and host your cloud services**.



- **Public cloud** > **fast to deploy** but with **less security**
- **Private cloud** > **Offers more control and security**
- **Hybrid cloud** > **combination of both public and private**

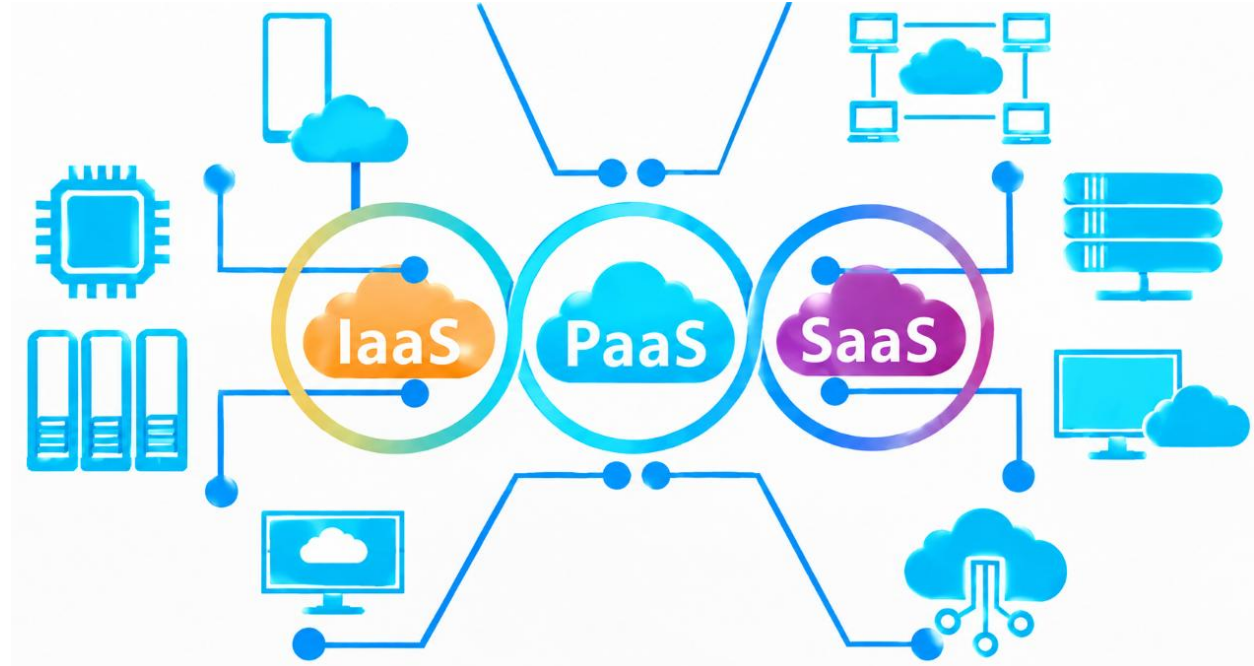
Why We Use Cloud Computing

1. **High Availability** – Keeps services **online and accessible**.
2. **Scalability** – Allows the **system to grow when needed**.
3. **Elasticity** – **Automatically grows or shrinks** resources **based on demand**.
4. **Agility** – Lets us **deploy resources quickly**.
5. **Reliability** – Helps the system **recover from failures**.
6. **Fault Tolerance** – Keeps the system **working even when something fails**.
7. **Global Reach** – Places services **closer to customers worldwide**.

8. **Measured Service** – **Pay only for what you use**. (pay per hour, pay per GB, pay per services)

Cloud Service Types

Define how much responsibility users manage versus what the provider manages

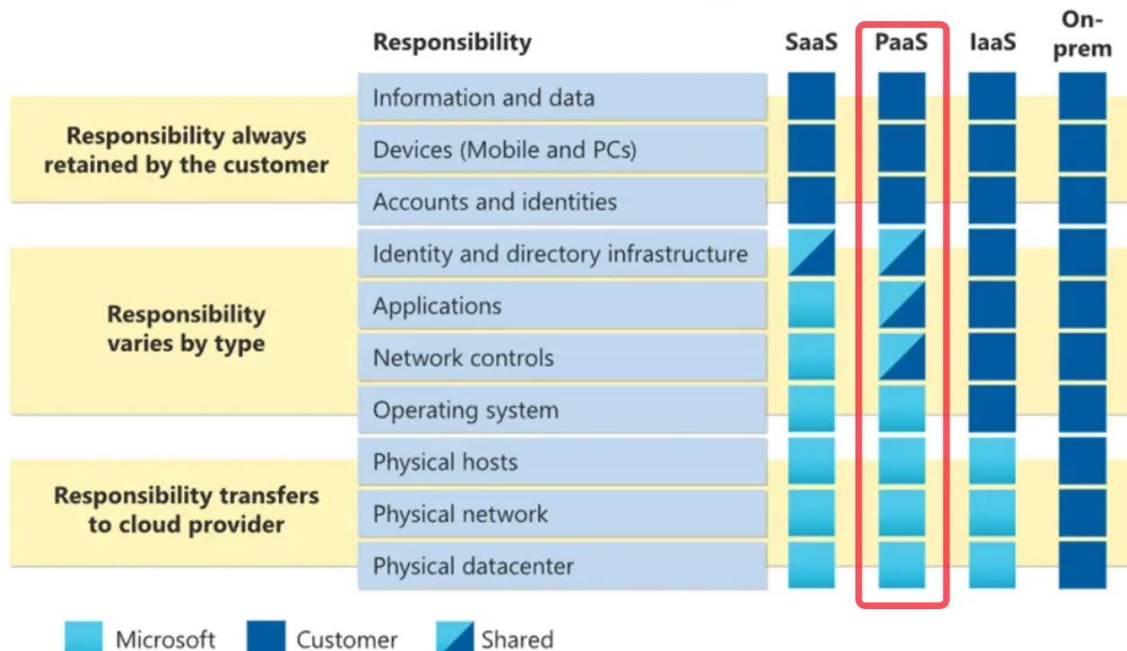


1. **IaaS – Infrastructure as a Service**
 - **You manage more**; cloud gives you the infrastructure.
 - Examples: **Azure Virtual Machines, Virtual Networks, Load Balancers**
2. **PaaS – Platform as a Service**
 - **Cloud manages the platform**; you focus on developing.
 - Examples: **Azure App Service, Azure SQL Database, Azure Functions**
3. **SaaS – Software as a Service**
 - **Everything is managed for you**; just use the software.
 - Examples: **Microsoft 365, Microsoft Defender, Microsoft Entra ID**

IaaS	Renting a car
PaaS	Hiring a driver
SaaS	Taking an Uber

Shared Responsibility Model

Defines who is accountable for security, compliance, and operations in the cloud.



- Microsoft always manages physical data centers, physical network, and physical hosts
- Users always manages data, identity and access control, endpoint security, and accounts.

Consumption based and Pricing Models (Capex vs OpEx)

Resources like virtual machines, storage, and bandwidth are metered.

Capex (capital expenditure)

A large **upfront costs for capitals** such as servers, software licenses and Data center.

OpEx (operational expenditure)

Paying for services as users consume them.

- Azure uses consumption -based on pricing- you only pay for what you use
- The cloud shifts spending from CapEx to OpEx, moving from large upfront costs with CapEx to flexible monthly billing with OpEx
- Azure offers reserved instances and spot **VWs** to reduce costs

SLA (Service Level Agreements)

Microsoft's formal promises of uptime for each Azure service

- common SLA percentages include 99.9% , 99.95%, 99.99% and 99.999%
- The higher the personage, the less downtime likely to experience
- When combining services, composite SLAs are calculated by multiplying individual service SLAs

Ex:

Question 5:

An organization deploys a web application using Azure App Service with a 99.95% SLA and connects it to Azure SQL Database with a 99.99% SLA.

☐ 99.99%

☐ 99.95%

☐ 99.97%

☒ 99.94%

Correct ✓

OBJ. 1.2 - Composite SLA is calculated by multiplying the individual SLA percentages of all dependent services in a solution. When combining Azure App Service at 99.95% with Azure SQL Database at 99.99%, the calculation is $0.9995 \times 0.9999 = 0.9994$, which equals 99.94%. The composite SLA is always lower than any individual service SLA because each dependency introduces additional points of potential failure. You cannot average or select the highest/lowest individual SLA, you must multiply all percentages together to determine overall system availability.

✚ Explain this further

Real-World SLA Example: E-Commerce Store on Azure App Service



- 2 nines, or 99% uptime = about 87.7 hr. of downtime per year
- 3 nines, or 99.9% = around 8.76 hours per year
- 99.95% = roughly 4.38 hrs. per year
- 4 nines or 99.99% = just under 53 min per year

Section 3

Regions, and Available Zones

Azure Region: A geographic area containing multiple datacenters.

Region effects

1. Performance
2. Availability
3. Cost
4. Compliance

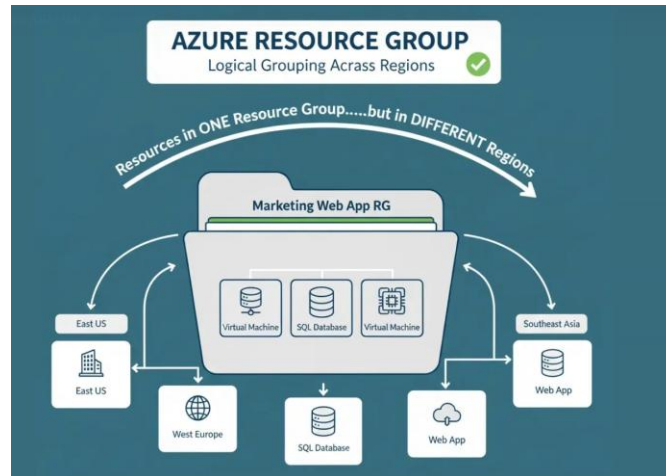
Azure has 60+ regions World Wide

Microsoft Azure services that support Zone Redundancy:

- Virtual Machines
- Storage
- Load Balancers
- **Availability Zone:** Separate datacenters with their own power, cooling, and networking for better availability.
- **Region Pair:** Two Azure regions connected for backup and disaster recovery.
- **Zone Redundancy:** Services like ZRS Storage and Azure SQL can run across zones for better availability.
- **Cost & Services:** Prices and available services can differ by region.
- **Data Transfer:** Moving data between regions may cost extra.
- **Sovereign Clouds:** Special isolated Azure environments for government, security, and legal requirements. Examples: Azure Government and Azure China.

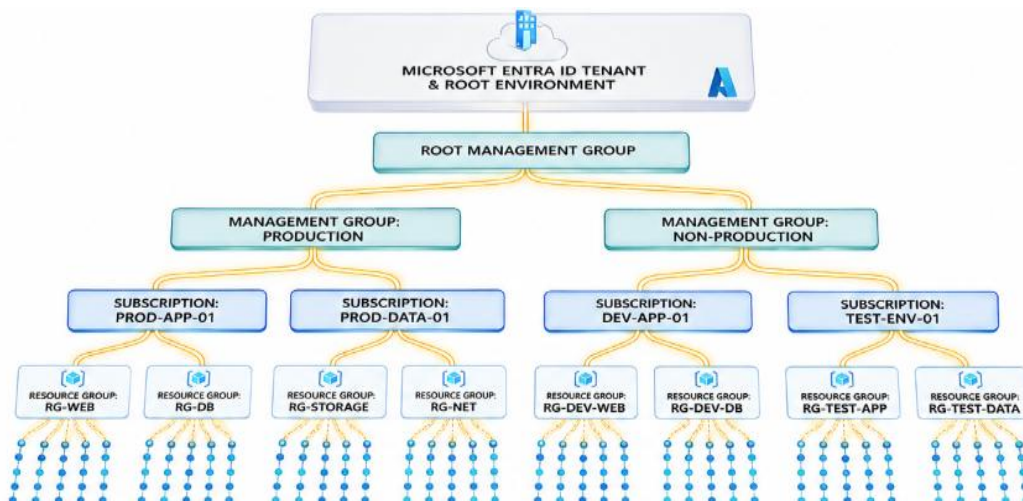
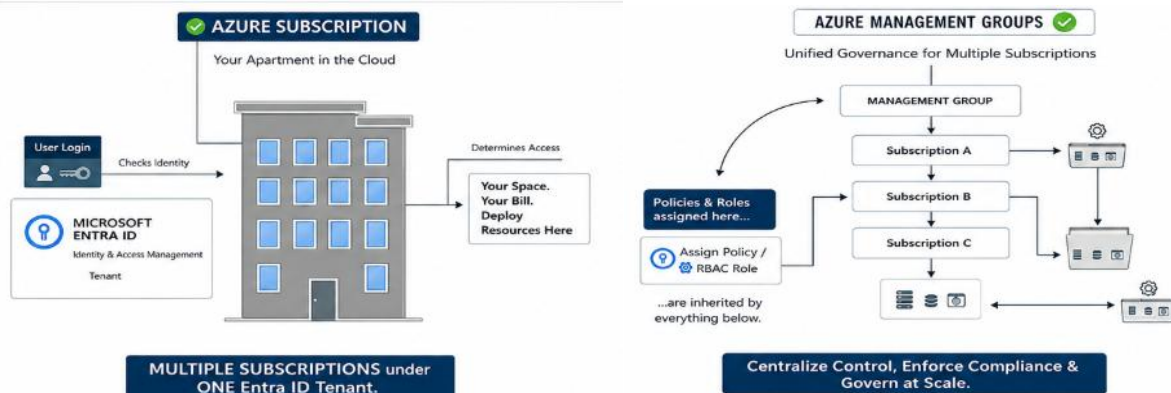
Resources and Resource Groups

- **Resources** — the deployable services in Azure, such as Virtual Machines, Storage Accounts, SQL Databases, and Virtual Networks.
- **Resource Groups (RGs)** — Logical containers used to manage related Azure resources.
- Resources in one Resource Group can be in different Azure regions. A **Resource Group has its own location**, but this location is mainly where Azure stores the **Resource Group's metadata**, not necessarily where its resources are located.
- **Tags are NOT inherited** — if you add a tag to a Resource Group, resources inside it **do not automatically receive that tag**.
- **RBAC roles are inherited** — Permissions assigned at the Resource Group level generally apply to the **resources inside it**. (Role Based Access Control)
- **Azure Policies apply to resources within their scope** — a policy assigned to a Resource Group can enforce rules on resources inside that group.
- **Deleting a Resource Group deletes its resources** — be very careful when deleting an RG.
- **Name convention** Rg-app-envy-location (us-dev-envy-va)



Subscriptions

Azure Subscription = the main billing and access boundary in Azure.



- You need a **subscription before you can deploy resources.**
- A subscription is associated with a **Microsoft Entra ID tenant**, which provides **identity, authentication, and access management.**

- **Management Groups** sit **above subscriptions** and allow organizations to manage **RBAC and Azure Policy** across multiple subscriptions.
- **Up to 6 levels:** Management Groups can be organized up to **6 levels deep**.
- **Root Management Group:** Azure **automatically creates a root Management Group when an Extra ID tenant is created**.
- **RBAC & Policies:** Permissions (RBAC) and Azure Policies can be set at any level. Lower levels inherit those settings.
- **Multiple subscriptions:** Large companies often use multiple Azure subscriptions to:
 - **Separate environments**, such as **Development, Testing, and Production**
 - **Track costs separately**
 - **Separate different departments or business units**
- **Simple hierarchy:**
- **Management Group → Subscription → Resource Group → Resources**
- **MSP = Managed Service Provider**
- MSPs are companies that **manage IT or cloud services for other companies**.
- They can **use Management Groups to organize and manage multiple customers** more easily.
- This allows them **to apply policies, permissions, and rules** across customer environments.

Azure Resource Manager (ARM)

- **ARM = Azure Resource Manager**
- **ARM** is the core **management layer of Azure**.
- ARM helps you **create, manage, organize, update, and delete** Azure resources.
- Resources can include **Virtual Machines, Storage Accounts, Databases, Networks**, etc.
- Simple definition:
- ARM is the system Azure uses to manage all your Azure resources in one consistent way.
- **Example: You want to create a Virtual Machine → ARM processes your request → Azure creates the VM.**

Infrastructure as Code (IaC) enables repeatable, automated, and versioned deployments

Templates can be deployed via Portal, CLI, PowerShell, or automated pipelines

General Data Protection Regulation (GDPR)

Used: **Rules that protect people's personal data**

Section 4

VM building Blocks:

Are the **main components and settings** use to create and configure a **Virtual Machine (VM)** in Azure?

Pricing based on:

- **VM Size: Determines the VM's CPU, RAM, and performance.** Bigger VM = higher cost.
- **Image: A template used to create a VM,** including the operating system (Windows/Linux) and sometimes preinstalled software.
- **Single VM: One virtual machine running by itself → 99.9% uptime.**
- **Availability Set: Keeps multiple VMs separated across different hardware** within the **same datacenter** to reduce hardware-related downtime → **99.95% uptime.**
- **Availability Zones: Places VMs in separate physical datacenters within the same Azure region → 99.99% uptime.**
- **Region: The geographic location where Azure resources run,** such as East US or West Europe.
- **Storage: The disk space to store VM's operating system, applications, and data.**
 - **HDD (Hard Disk Drive): Slower, but cheaper.**
 - **SSD (Solid-State Drive): Faster and good performance, more expensive.**

**VMs = Full OS + Full Control = IaaS
(Infrastructure as a Service)**

Role-Based Access Control (RBAC)

- **RBAC controls who can access Azure resources** and what they can do.
- ARM allows RBAC permissions to be applied at different levels:

Management Group → Subscription → Resource Group → Resource

VMSS, Availability Sets and Azure virtual desktop

- **VMSS (Virtual Machine Scale Sets): Automatically adds or removes VMs based on demand.** Used for scaling.
- **Availability Set: Distributes VMs across different hardware to protect against hardware and maintenance failures.**
- **Availability Zones: Runs VMs across separate datacenters within the same Azure region** for higher availability.
- **Azure Virtual Desktop: Provides a Windows desktop and applications through the cloud,** accessible remotely.
- **Azure Arc: use to manage multi-cloud, and Azure resources** using Azure management tools.

- **Availability Sets/Zones:** Used for **redundancy and higher uptime**.
- **SLA (Service Level Agreement):** Microsoft's **uptime commitment** for an **Azure service**.

Vertical = More power per machine (requires downtime)

Horizontal = More machines (no downtime, handled by VMSS)



VMs Types

Means the **different kinds or sizes of VMs (Virtual Machines)** available in Azure.

- **Choose VM size based on workload, CPU, memory, or storage-intensive.**
- **Storage options:** Standard HDD (cheap), Standard SSD (balanced), Premium SSD (fastest, required for 99.9% SLA).
- **Larger VM sizes and higher-performance disks cost more.**
- Use **Azure Advisor** for optimization tips.
- Use **Reserved Instances** for cost savings on predictable workloads, up to 72% for 3-year commitments.

B	Budget (burstable)
D	Daily use (general-purpose)
E	Elephant memory (RAM-heavy)
F	Fast CPU (compute-optimized)
M	Massive memory (enterprise-scale)

App Hosting Options

- **Azure app service** -> **A Platform** as a Service (PaaS) designed specifically **for hosting web applications and APIs**.
 - **NET**
 - **Node.js**
 - **Python**
 - **Java**
 - **PHP**

Plan Tiers:

- **Free/Shared:** For Development and testing
- **Basic:** For dedicated compute instances, but no auto-scaling or deployment slots
- **Standard:** Can get auto-scaling up to 10 instances, deployment slots for staging, and daily backups.
- **Premium:** offers enhanced performance, up to 30 instances, and additional scaling options.
- **Isolated:** Runs app in a dedicated Azure Virtual Network, providing network isolation and maximum scale.



Deployment slots require Standard tier or higher and enable zero-downtime deployments

- **App Service includes built-in load balancing, auto-scaling, HTTPS, and CI/CD.**
- **Choose App Service when you want managed hosting, automatic scaling, and minimal operational overhead.**
- **Choose VMs when you need OS-level control or legacy app support.**
- **Choose containers for portability and micro services orchestration.**

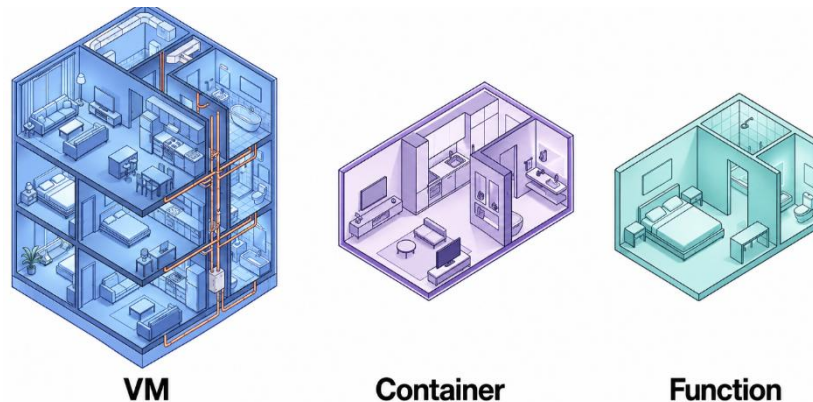
VMs, containers, functions difference

- **VM (Virtual Machine): A full virtual computer** with its own OS, CPU, memory, and storage.

- **Container:** A **lightweight package** containing an application + its dependencies, without needing a full separate OS.

Two Main Azure Container Services:

- **ACI (Azure Container Instances):** Used to **quickly run simple containers** without managing servers.
- **AKS (Azure Kubernetes Service):** Used to **manage many containers** and provides features such as scaling and load balancing.



Easy to remember:

- **VM = Apartment**
- **Container = Studio**
- **Function = Hotel room**

Section 5

Virtual Networks and subnets

- **Virtual Networks (VNETs)** are **private cloud networks in Azure**.
- **Subnets** are IP addresses that **define a person's computer**
- **Vnet Peering:** Connects two separate Azure Virtual Networks (Vnets) using Azure's private backbone.
- **Peering is non-transitive by default**, so plan your peering topology carefully.
- **Remember:** if VNet A peers with B, and B peers with C, A and C cannot communicate unless you create a direct peering between them.
- **NSG (Network Security Group):** Controls which **network traffic** is allowed or denied. **Firewall**
- **UDR (User-Defined Route):** Lets you **create custom routing rules** for network traffic. **Router**
- You can apply custom routing and security rules as needed with options like UDRs and NSGs.
- **After the slash (/),** the smaller the number, the more addresses; the bigger the number, the smaller addresses.

Azure DNS Domain name Services

- **Azure DNS hosts** DNS zones and records in Azure.
- You can manage domains like "**diontraining.com**" directly in Azure without separate DNS servers.
- **Public DNS zones** handle internet-facing domains.
- **Private DNS zones** handle internal name resolution for Azure VNets.
- **Service DNS zones** simplify hybrid network management.
- **Auto-registration** automatically creates DNS records for new VMs.

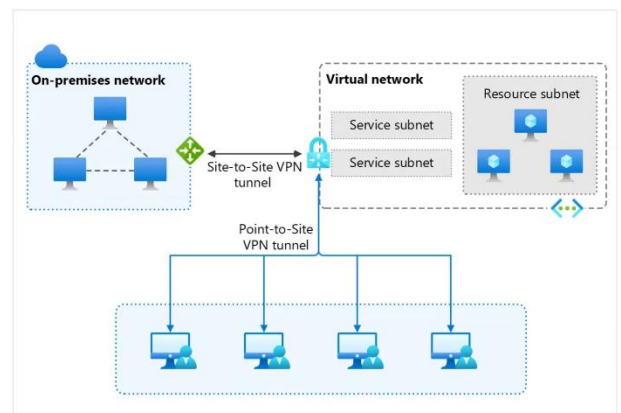
A	Maps names to IPv4 addresses
CNAME	Creates aliases for other domain names
MX	Defines mail servers for email delivery
TXT	Stores text for verification and other purposes

Azure Public, Private, and Service Endpoints

- **Public Endpoint:** **Exposes resources to the internet using public IP addresses.**
 - Used for services that must be accessible from the internet.
 - Use only when internet access is required.
- **Private Endpoint:** **Assigns a private IP within a VNet** and keeps traffic off the public internet.
 - Used to eliminate internet exposure.
 - Best for security and compliance.
- **Service Endpoint:** **Extends VNet access to Azure services** while using the Microsoft backbone.
 - Provides secure access to Azure services without DNS changes or private IP management.
 - Best for secure, high-performance access.

VPN Gateway, Bastion, and Express Route

- **VPN Gateway** provides secure connections over the public internet using encryption
- **Site-to-Site(S2S):** for networks
- **Point-to-Site(P2S):** for individual users
- **Azure Bastion:** allows secure RDP/SSH access to Azure VMs without public IPs
- **RDP (Remote Desktop Protocol):** Used to remotely connect to and control a Windows computer or VM using a graphical desktop.
- **SSH (Secure Shell):** used to securely connect to and manage a Linux computer or VM using command-line commands.
- **ExpressRoute:** creates a private, high-speed connection between on-premises network and Azure
- **Redundancy:** having a backup connection in case something fails



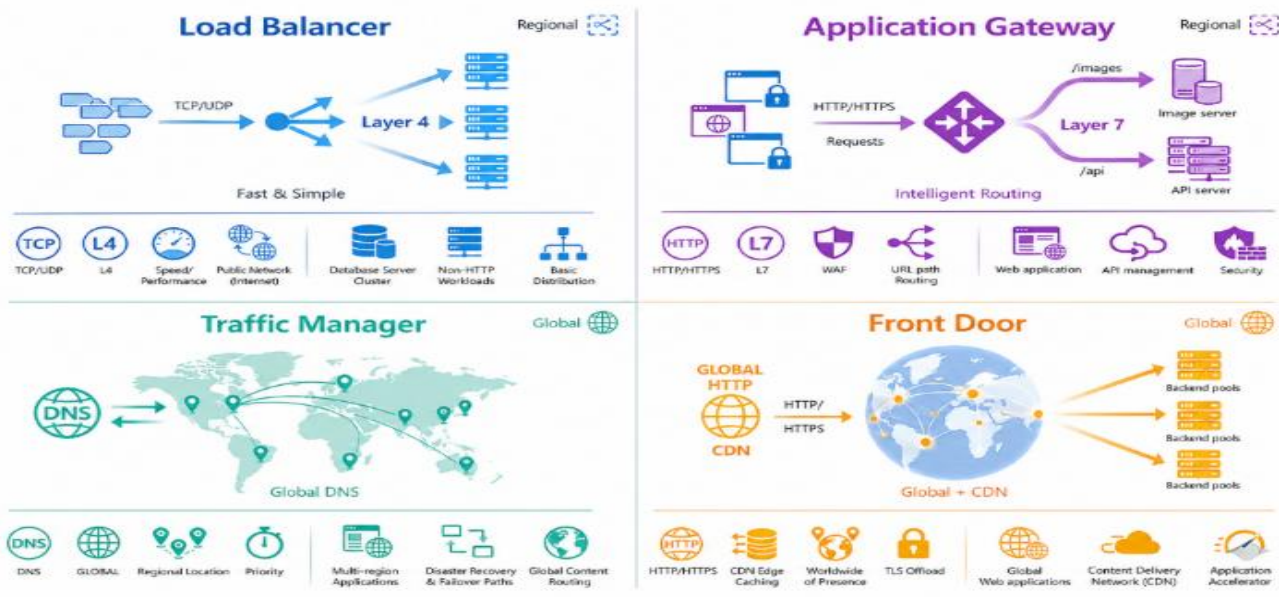
Load Balancing Services

- **Azure Load Balancer:**
Handles TCP/UDP network traffic, both inbound and outbound
- **Azure Application Gateways:**
Inspect HTTP and HTTPS content and make routing decisions based on URL paths, headers, cookies, and more
- **Azure Traffic Manager:**
DNS-based service that sends users to the best available server.
- **Azure Front Door:**
Sends web traffic to the best server and provides speed and security.

Network and Application Security Groups

- **Network Security Groups (NSGs)** a set of rules that control inbound and outbound traffic to azure resources.
- **Application Security Groups (ASGs)** let users group VMs by application role.
- They use priority-based rules and are evaluated in order from lowest to highest priority number.
- They deny inbound traffic from the internet by default and can be applied to subnets or NICs.

- Users can reference ASGs in NSG rules for better management and scalability.
- NSGs work at Layer 3 and 4.
- Azure Firewall works from Layer 3 to Layer 7 and provides centralized, stateful inspection across your entire Azure environment.
- Use Network Watcher to troubleshoot rules effectively and verify connectivity.
- Use Azure DDoS Protection alongside NSGs for protection against volumetric attacks.

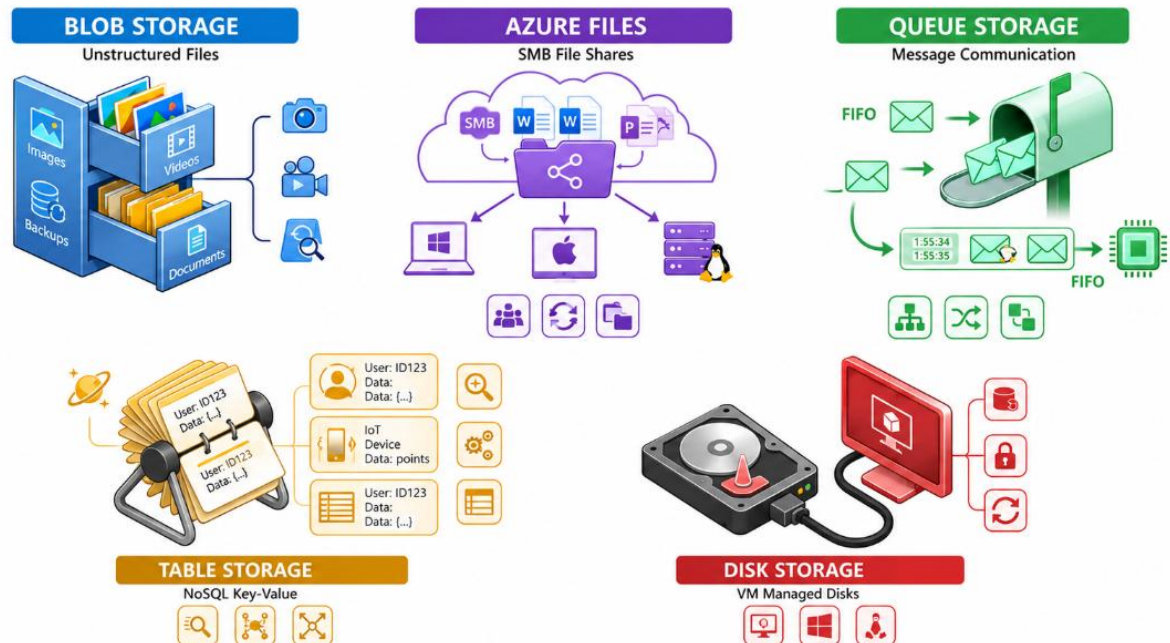


Section 6

Storage Account Types and Services

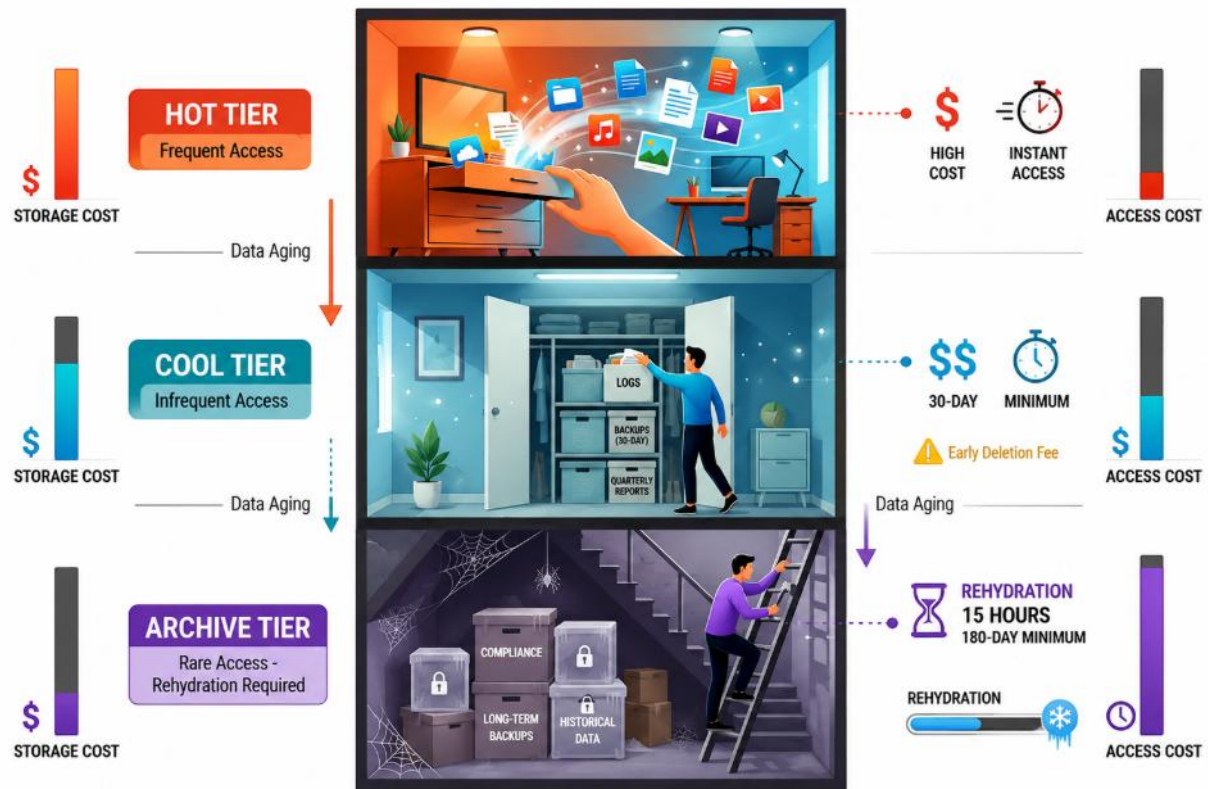
- **Blob storage:** unstructured files like images and videos.
- **Azure files** provides SMB files shares accessible across the world
- **Queue stores** messages for app communication
- **Table storage:** table stores NoSQL structures data (being superseded by cosmos DB for table)
- **Disk storage:** disk storage provides manages disks and VMs
Preformed tiers include:
 - **Standard:** Lower cost, generally lower performance.

- **Premium:** Higher cost and higher performance, typically SSD-based.



Storage Tiers: Hot, Cool, and Archive

- **Hot Tier** -> is for **frequent** access and fast performance, highest storage cost, lowest access cost.
- **Cool tier** -> is for **infrequent** access and lower storage cost, higher access cost, 30-day minimum storage duration.
- **Archive** -> is for **rarely** accessed data with the lowest storage cost but retrieval delays, 180-day minimum storage duration.
- The Archive tier requires rehydration before use, and that comes with time and cost standard rehydration takes up 15 hours.



Redundancy Options

• LRS – Locally Redundant Storage

- **3 copies in one datacenter.**
- Lowest cost.
- 11 nines durability.
- **Protects against server/rack failure.**

• ZRS – Zone-Redundant Storage

- **3 copies across Availability Zones.**
- **Better protection.**
- 12 nines durability.

• GRS – Geo-Redundant Storage

- 6 total copies: **3 local + 3 in a paired region.**
- **Adds disaster recovery through a secondary region.**
- 16 nines durability.

• RA-GRS – Read-Access Geo-Redundant Storage

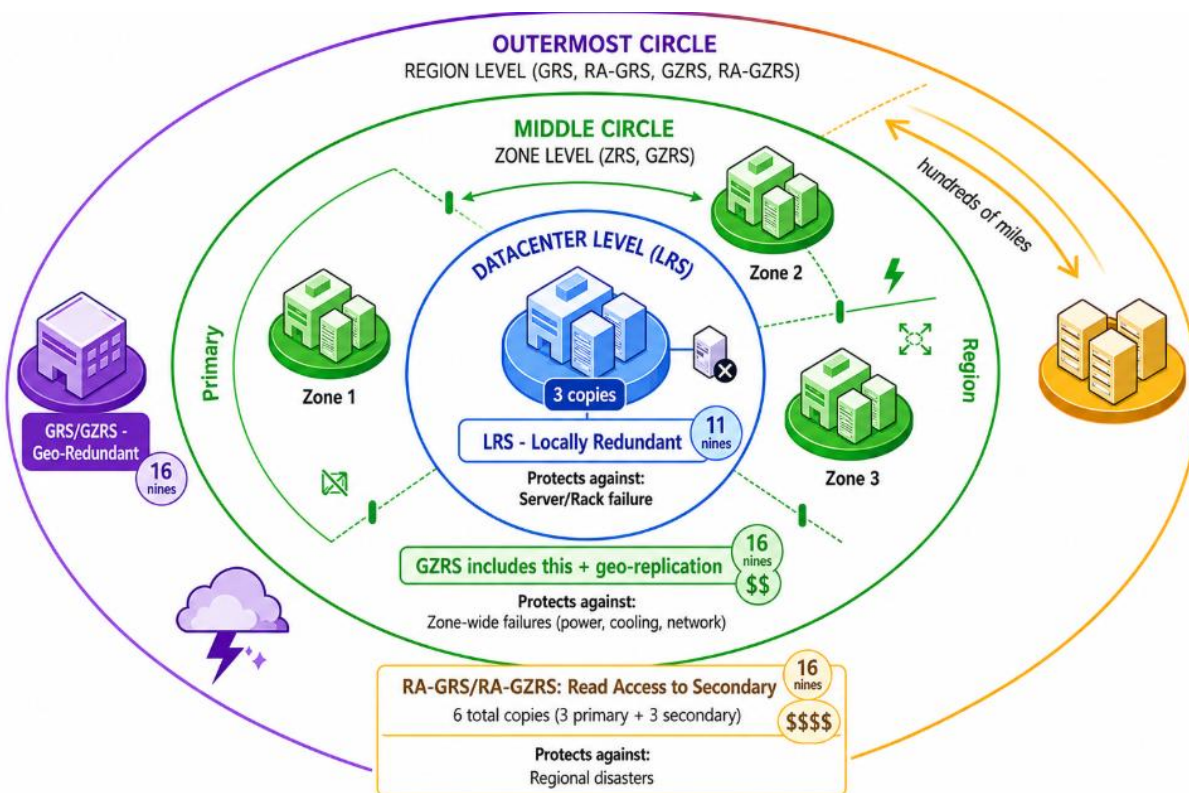
- Same as GRS, but **add read access to the secondary copy.**
- Good for read-heavy, globally distributed apps.

• GZRS – Geo-Zone-Redundant Storage

- Combines **ZRS in the primary region** with geo-replication to a secondary region.
- **Provides both zone and regional protection.**

• RA-GZRS – Read-Access Geo-Zone-Redundant Storage

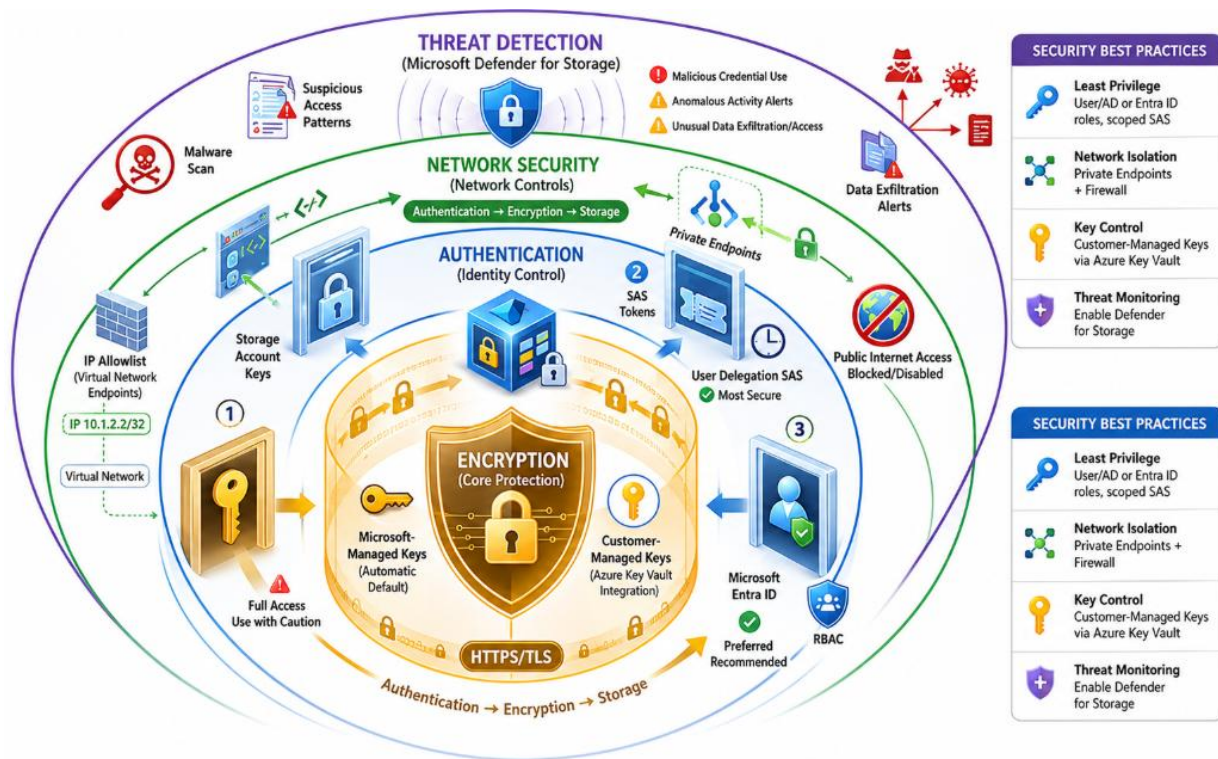
- Same as **GZRS + read access to the secondary region.**
- **Maximum protection and availability.**



Secure Storage Access

- Azure offers three access methods: **Storage Account Keys, SAS, and Microsoft Entra ID.**
- **Storage Account Keys** = **full access**. Should be used sparingly and rotated regularly.
- **SAS (Shared Access Signature)** = **scoped, time-limited access** for delegated use. User Delegation SAS is most secure because it uses Microsoft Entra ID credentials.

- **Microsoft Entra ID** = **secure, role-based access tied to identity**; preferred for enterprise scenarios.
- **Encryption at rest** is automatic; choose between Microsoft-managed keys and customer-managed keys via Azure Key Vault.
- **Encryption in transit** uses HTTPS/TLS. Can enforce secure transfer in storage account settings.
- **Storage firewalls** restrict access to specific virtual networks or IP ranges.
- **Private endpoints** help isolate storage traffic from the public internet by giving storage accounts private IPs within the virtual network.
- **Microsoft Defender for Storage detects threats** like malware uploads and suspicious access patterns.



File Transfer Tools

- Azure offers several tools to transfer files into storage.
- **AZ Copy** is **command-line** based and best for **automation, bulk transfers**, and scenarios requiring **high performance**.
- **Storage Explorer** is **GUI-based** and great for **managing data visually**.
- **Azure Portal** allows **manual, small-scale uploads** through a browser.
- **Azure File Sync** keeps **on-premises Windows** file servers **synchronized** with **Azure Files**.

- **Azure Data Box** handles **offline, large-volume migrations** when network transfer isn't practical.

Easy way to remember:

- **AZ copy** -> for speed and automation
- **Storage explorer** -> for GUI experience
- **Azure portal** -> for quick, one off uploads
- **Azure file sync** -> for hybrid file server scenarios and synchronization

Example of AZ copy command:

```
azcopy copy 'C:\local\folder\*'
'https://mystorageaccount.blob.core.windows.net/mycontainer' --recursive
```

Data Migration Tools

The process of moving data or workloads from on-premises environment into Azure.

Azure Migrate

Go-to tool for assessing, planning, and migrating virtual machines, databases, web apps, and servers into Azure.

- Migration hub for assessment and workload migration
- Best for VMs, databases, and web apps
- Includes dependency mapping, cost estimation, Server Assessment, Server Migration, Database Migration, and Web App Migration.

Azure Import/Export

- Ship users' own drives for **offline data transfer**
- Cost-effective when you have existing hardware

Azure Data Box

- Offline **physical transfer of large data** volumes
- Three options: Disk (40TB), standard (80TB), Heavy (800TB)
- Perfect for limited bandwidth scenarios

Azure File Sync

- Synchronizations between on-premises Windows file servers and Azure Files
- Supports cloud tiering and hybrid access
- Part of both transfer and migration strategies

Section 7

Microsoft Entra ID and domain services

Microsoft Entra ID

- Microsoft Entra ID is a **cloud-based identity and access service**.
- Microsoft Entra ID is different from on-prem Active Directory.
- Entra Domain Services brings legacy protocols to the cloud with managed domain controllers.
- Entra ID tenants **store identities**.
- Microsoft Entra Connect enables hybrid identity by syncing on-prem users to the cloud.

Single Sign-On

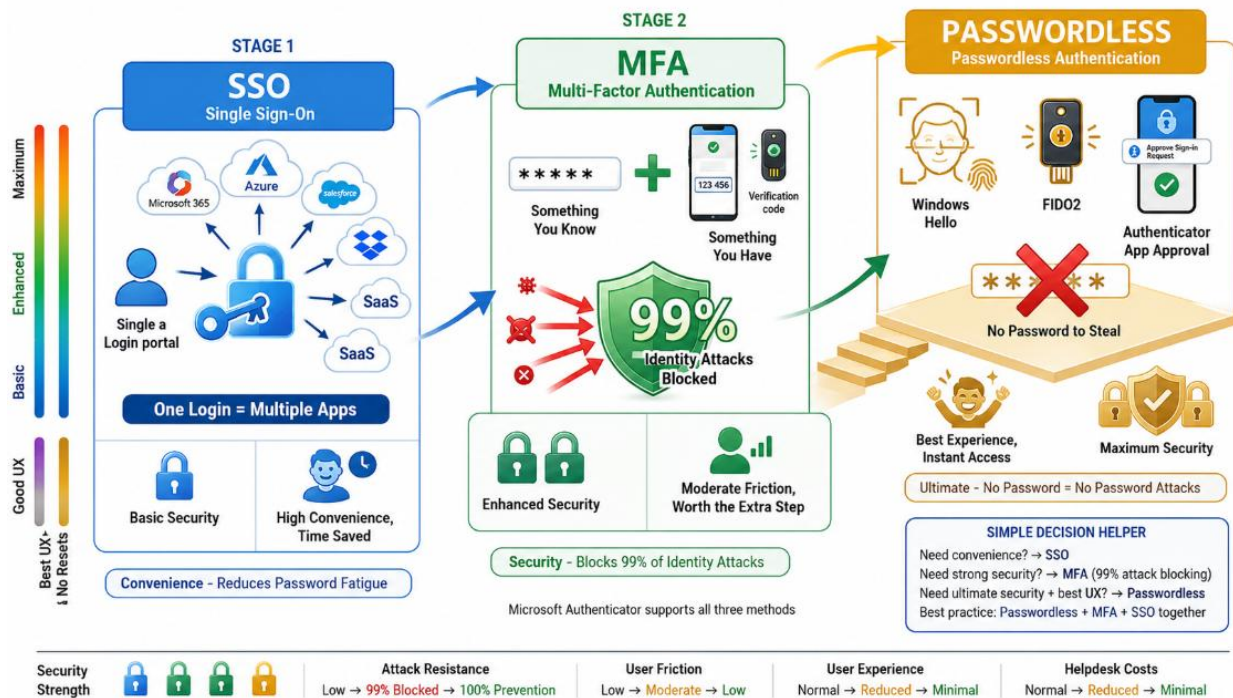
- Microsoft Entra ID uses protocols like **SAML and OpenID Connect** to let users access apps securely. Ms.eID gives you **secure access to apps**.
- Once the user authenticates with Entra ID, that token is used to **grant access to connected services**.
- Users can configure SSO for **Microsoft apps, Azure services, and thousands of third-party SaaS platforms like Salesforce and Dropbox**.

Multi-factor Authentication

- **MFA is verification of** Text message code, phone call, mobile app notification, or app-generated code.
- MFA can be enforced using **Conditional Access policies**.
- Conditional Access is how Azure evaluates the **login context and enforces policies like requiring MFA**.
- MFA is included in all Microsoft Entra ID editions, including the **free tier**.

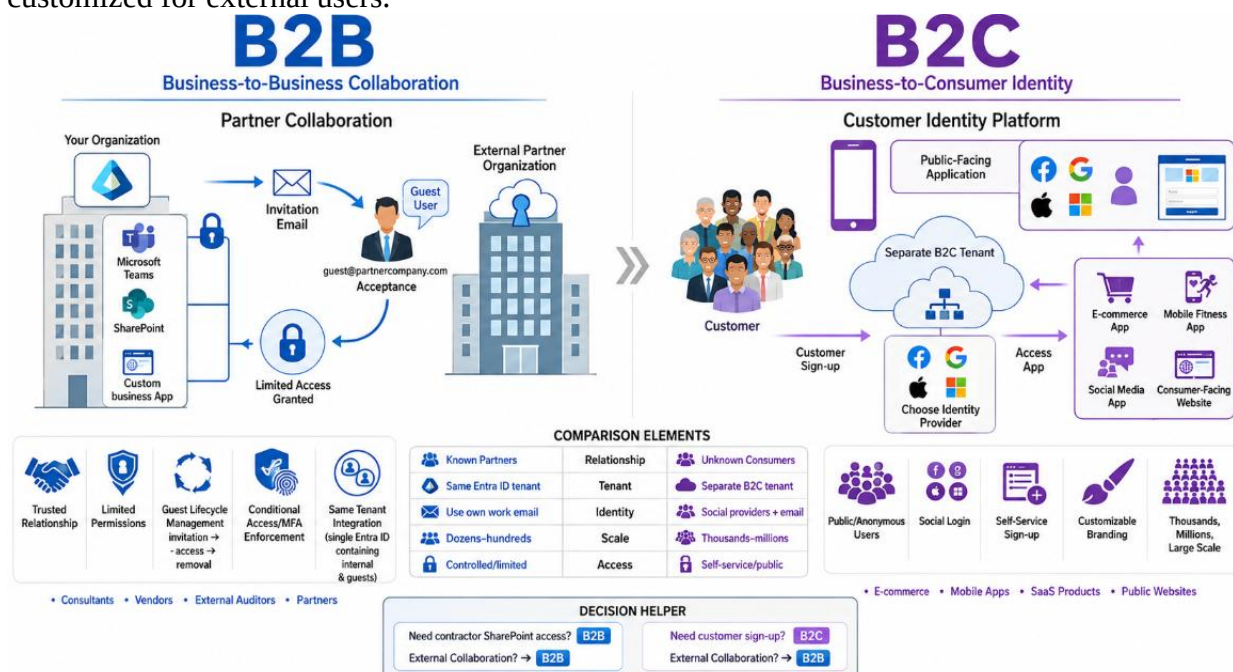
Key points:

- **MFA (Multi-Factor Authentication) equals password plus something else**. Blocks over 99% of identity attacks.
- **SSO (Single Sign-On) equals one sign-in, multiple app access**.
- **Password less equals no passwords, more security and better experience**.
- **Conditional Access** is how you enforce MFA with granular control.



B2B and B2C

- **B2B** supports business-to-business collaboration with **guest access to applications**.
- **B2C** is a business-to-consumer identity platform for public applications.
- **B2B** is integrated with the Microsoft Entra ID tenant; **B2C** is a separate, dedicated tenant, customized for external users.



Role-Based Access control (RBAC)

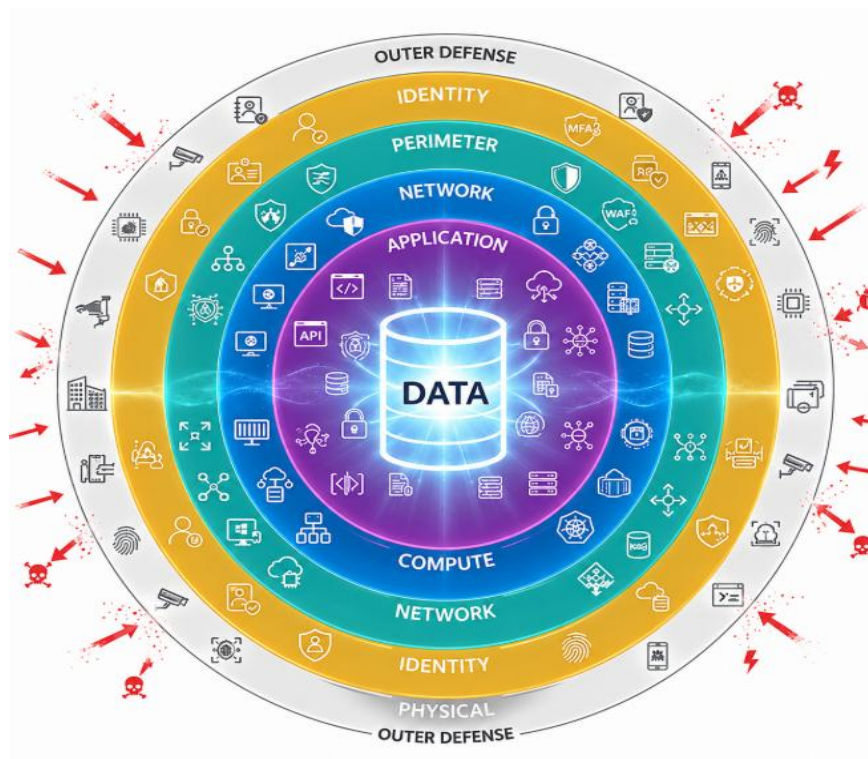
The system Azure uses to manage permissions; it defines what users can do within Azure based on their assigned role.

Zero Trust & conditional access

- **Zero Trust** equals "**Never trust, always verify.**"
- Three principles: **verify explicitly, use least privilege, and assume breach.**
- **Conditional Access** equals real-time access decisions based on signals.
- Conditional Access enforces policies like **requiring MFA or blocking untrusted logins.**
- Common use case: **Require MFA for users signing in outside of trusted networks.**
- Conditional Access is managed in **Microsoft Entra admin center.**
- Conditional Access integrates with **Microsoft Entra ID Protection** for risk-based policies and **Microsoft Intune** for device compliance.

Defense in depth

- **Defense in Depth** equals multiple security layers, also called **layered security.**
- The seven layers are: **Physical, Identity, Perimeter, Network, Compute, Application, and Data.**
- **Microsoft Defender for Cloud** is the security center.
- **Secure Score** tells how secure an environment is from a percentage of **0 to 100.**
- The **Compliance Dashboard** maps setup to real-world regulations.
- **Azure Policy** works behind the scenes to evaluate resources.
- **Defender** turns policy results into action items with estimated implementation costs.
- **Defender for Cloud** protects multicloud and hybrid environments, including **Azure, on-premises, AWS, and GCP.**



Microsoft defender for cloud

- **Microsoft Defender for Cloud** equals **centralized security management tool**.
- **Secure Score** equals your security grade, **0 to 100%**.
- **Recommendations** equal steps to improve security, based on **Azure Policy** and **Azure Security Benchmark**.
- **Threat protection** provides alerting and monitoring for attacks across supported workloads.
- **Compliance** shows alignment with standards like **ISO 27001, NIST, PCI DSS, and GDPR**.
- **Free tier** equals foundational **CSPM** with posture management and Azure security recommendations.
- **Paid Defender plans** equal enhanced **CWPP** with threat protection for specific workloads.
- **Defender for Cloud** integrates with **Azure Policy** for compliance and **Microsoft Sentinel** for advanced threat response.

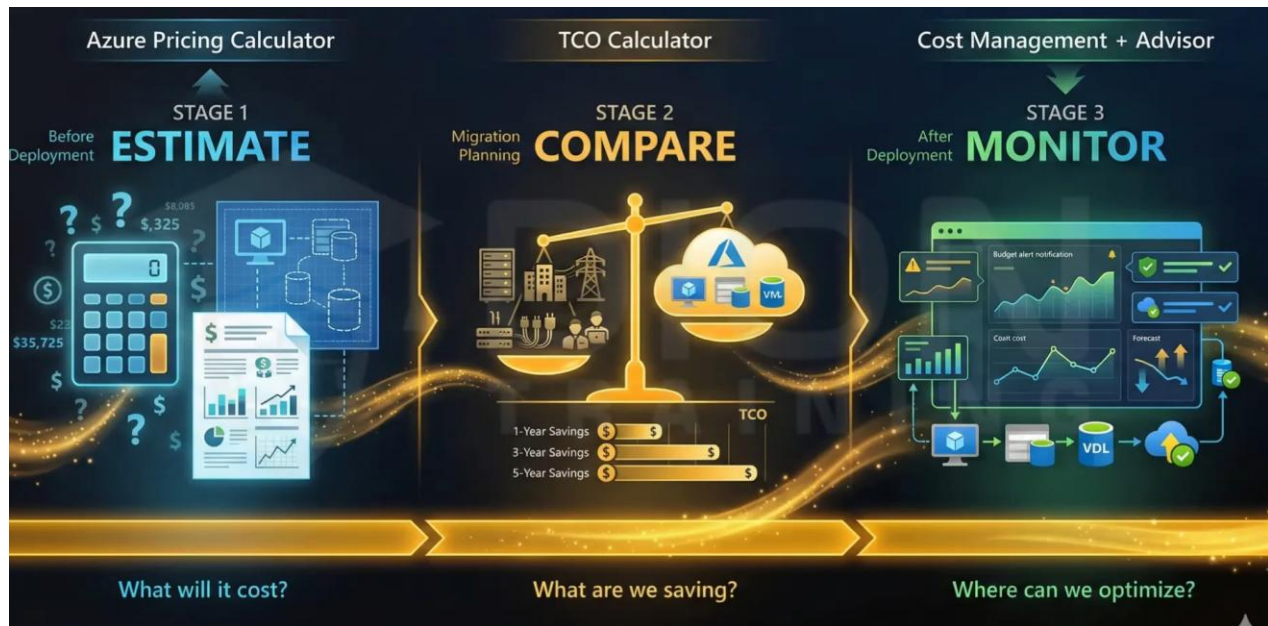
Section 8

Control and Optimize Azure Costs

- **Cloud costs increase when resources are left running**, overprovisioned, data transfer is ignored, or tagging is not used.
- **Azure Cost Management + Billing** is a free **toolkit used to monitor and control spending**.
- **Cost Analysis** shows where spending occurs and forecasts future costs.
- **Budgets** set spending limits and define thresholds for alerts.
- **Alerts sent through Action Groups** notify when a budget threshold is being approached or exceeded.
- **Azure Advisor** gives data-driven recommendations to save money.
- **Tags** provide cost attribution, but they must be applied consistently at the resource level.
- **Budgets alert but do not enforce**.
- **Proactive strategies** such as **Reserved Instances**, **Azure Hybrid Benefit**, **shutting down dev test workloads**, and **right-sizing resources** reduce costs.

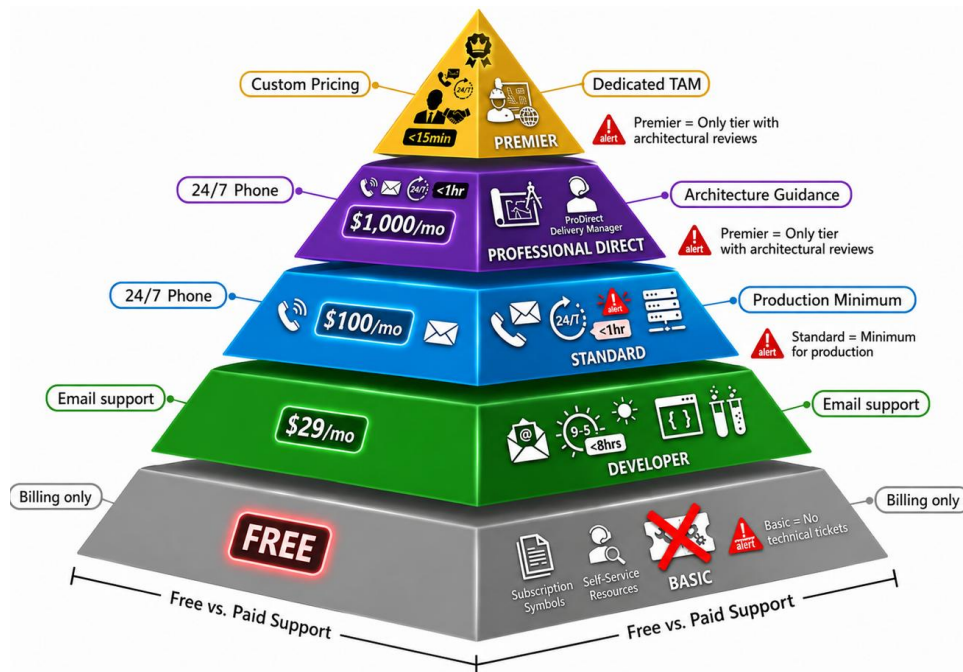
Azure Pricing Calculator and TCO Calculator

- **Azure price Calculator** show us our Azure future cost
- **TCO calculator** used for compare between different resource
- **The Azure Pricing Calculator** helps users estimate ongoing monthly costs before deploying resources.
- **Used for migration planning** by comparing the cost of running workloads in **Azure** versus **on-premises environments**.
- **The TCO Calculator** helps users compare Azure to on-premises costs over multiple years to justify migration.
- **Use the tools** to avoid surprise bills and plan more effectively.
- After deployment, **Azure Cost Management and Budgets** help monitor and control spending.
- **Cost Management** includes forecasting to predict future costs based on trends.
- **Azure Advisor** offers cost-saving suggestions based on the user's actual usage.



Azure Support Plans and Options

- **Azure offers five main support plans: Basic, Developer, Standard, Professional Direct, and Premier.**
- **Basic** is free but does not allow technical support tickets.
- **Developer** provides email support during business hours, perfect for non-production environments.
- **Standard and above** include phone support.
- **Professional Direct** adds architecture guidance and proactive support from delivery managers.
- **Premier** is the enterprise tier with dedicated resources and architectural reviews from Microsoft.
- **The higher the tier, the faster the response time and the deeper the technical support.**
- **Support plans are a cost factor.**



Resource Tags and Locks

- **Tags are key-value labels that help organize resources** and manage costs.
- They can be applied to **resources, resource groups, and subscriptions**, but are **not inherited**.
- **Tags** support automation, policy enforcement, and financial reporting, with a limit of **50 tags per resource**.
- **Resource Locks protect resources from accidental deletion** or modification.
 - **Use Read-Only locks** to prevent configuration changes while allowing data operations.
 - **Use Delete locks** to stop deletions only.
 - **Locks ARE inherited** from parent to child resources, unlike tags.



Azure Policy

- **Azure Policy** is a governance tool that enforces compliance automatically.
- It uses **Definitions and Assignments** to apply rules, with optional exclusions for flexibility.
- **Policy effects** include Deny, Audit, Allow, Append, Modify, AuditIfNotExists, and DeployIfNotExists.
- **Policies evaluate during resource creation** and run compliance scans every **24 hours** for existing resources.
- **Azure Policy** restricts **what can be done**, while **RBAC** controls **who can do it**.
- Use **Policy Initiatives** for broad compliance, like **PCI DSS**, but remember these are audit policies that report status.
- Use **Defender for Cloud** to monitor your compliance posture in **real-time**.

		RBAC	AZURE POLICY
Deny	Blocks the action completely	ACCESS CONTROL Role Assignment Permissions Resource Access	COMPLIANCE MANAGEMENT Policy Enforcement Auditing & Monitoring Governance
Audit	Logs the action but doesn't block it		
Allow	Lets the action go through if it meets the rule		
Append	Adds extra settings to the resource automatically		
Modify	Changes the request to match the policy		
AuditIfNotExists	Checks for related resources		
DeployIfNotExists	Automatically deploy missing configurations		

Microsoft Purview

- **Microsoft Purview** gives visibility and control over data estate.
- It **catalogs and classifies data** to detect sensitive content like PII.
- It tracks **data lineage** to show how data moves through the organization.
- **Purview** identifies sensitive data but relies on other tools like **Microsoft Information Protection** and **Azure Policy** for enforcement.
- It helps meet **compliance requirements** by providing **discovery, classification, and reporting capabilities**.

1	Scan storage accounts and databases
2	Discover hundreds of PII records scattered across multiple environments
3	Classify the data and identify where sensitive information exists
4	Use Microsoft Information Protection to apply encryption
5	Use Azure Policy to ensure new resources meet the standard



Section 9

Azure Portal

- The **Azure Portal** is a web-based **GUI** at portal.azure.com.
- **Where you can manage, monitor, and configure all Azure services** without installing anything.
- Benefits include customizable dashboards, built-in Cloud Shell, Activity Logs, and access to quick start templates.
- The Portal is resilient, always available, and updates continuously without downtime.
- It's ideal for hands-on tasks, but not suited for automation or bulk deployments.

Cloud Shell

- **Azure Cloud Shell is a browser-based terminal** available on **Azure Portal** and at shell.azure.com that runs **Linux commands which create and delete resource groups and tagging VMs**.
- It runs on Azure Linux and automatically authenticates with your Azure account.
- It supports Azure CLI and Azure PowerShell.
- CLI commands start with "az" and are great for Linux and cross-platform automation.
- PowerShell uses cmdlets like New-AzVM and fits best in Windows environments.
- Cloud Shell includes 5 GB of persistent storage for saving scripts and files.
- It also includes a built-in code editor for creating and modifying scripts.
- Sessions time out after 20 minutes of inactivity, but your files persist.
- Use Cloud Shell to automate tasks like creating and deleting resource groups or tagging VMs.

Azure Arc

- **Azure Arc lets you combine** on-premises and **multi-cloud resources into Azure** for unified management.
- You can onboard **servers, Kubernetes clusters, SQL Servers**, and even run **Azure data services anywhere**.
- It enables centralized **Azure Policy, Role-Based Access Control (RBAC), and monitoring** for external resources.
- **Arc-enabled resources** appear in the **Azure Portal** just like native Azure resources.
- **Azure Arc** simplifies **hybrid cloud governance** by using a **single management plane**.



Understanding infrastructure as a code IAC

Version Control

- Track changes over time
- Roll back any resources
- Enforce standards through code reviews

Infrastructure as Code (IaC)

- **Automates Azure deployments through code.**

- All IaC in Azure works through Azure Resource Manager (ARM).

ARM Templates

- Use **JSON** and are powerful but complex.

Bicep

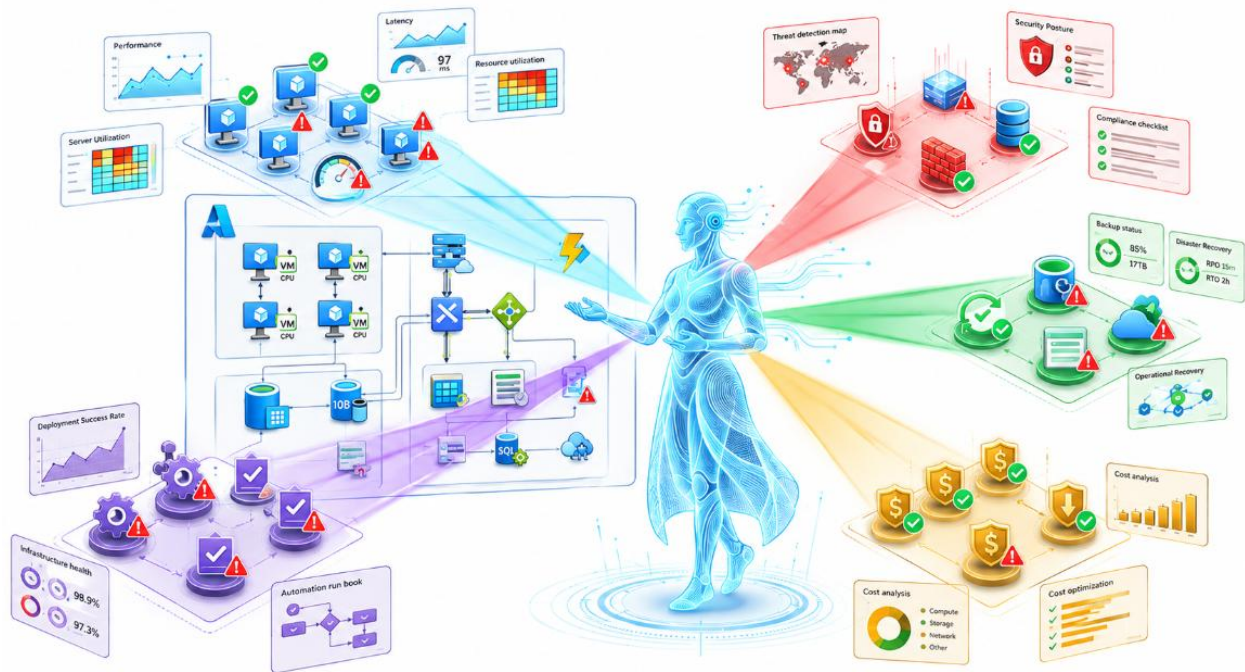
- Simplifies syntax and compiles down to ARM JSON.

Terraform

- Another option for cross-cloud environments and also uses ARM when deploying to Azure.

Azure advisor

- **Azure Advisor** is a built-in, **free Azure tool** that provides personalized recommendations.
- It analyzes telemetry and usage data from your resources.
- **It provides suggestions across** five categories:
 - **Performance**
 - **Security**
 - **Cost**
 - Reliability
 - Operational Excellence
- Cost recommendations include estimated savings amounts to help you prioritize actions.
- Cost recommendations help with proactive resource management, cost savings, and security hardening.
- Advisor integrates with other Azure tools like Cost Management and Defender for Cloud.



Service health, resource health, and global status

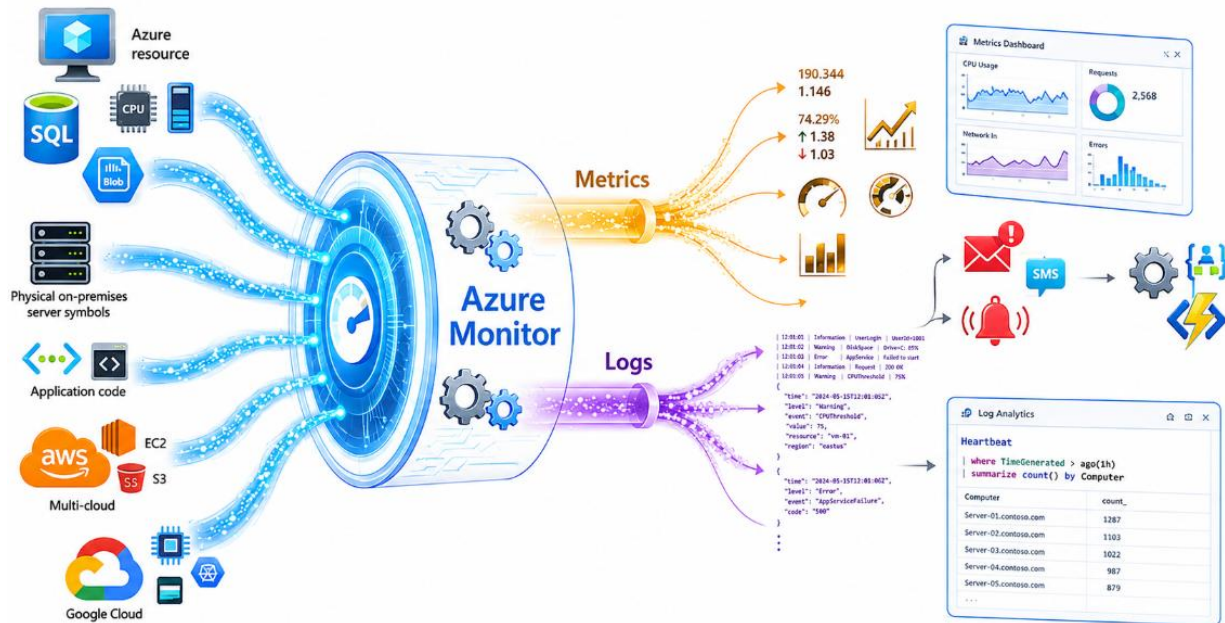
- **Azure Service Health:** Shows **service-level issues** relevant to the environment.
- **Resource Health:** Gives **resource-specific health data**.
- **Azure Status Page:** Shows the **global status of all Azure services**.

Service Health Alerts

- **Service Health alerts** can be configured through **Action Groups** for proactive notifications.
- **Service Health retains data for 90 days** for audit and compliance review.
- Alerts can trigger **automated responses** via **Azure Functions or Logic Apps**.

Azure monitor

- Azure **Monitors** collects, analyzes, and responds to **telemetry data** from Azure, on-premises, and **multi-cloud environments**.
- It handles two main data types: **metrics (numerical)** and **logs (event-based)**.
- **Log Analytics** is where logs are stored and uses **Kusto Query Language (KQL)** for powerful data analysis.
- **Azure Monitor** supports **alerts and automation**, helping users respond quickly to issues.



<https://aka.ms/AZ-900onYouTube>